

Causal Kernel Descent for Finite-Order Linear Recurrences

Abstract

We show that unnormalized causal kernel descent (CKD) can fail after state lifting: a bounded linear AR(2) example has a fixed-point prediction residual converging to 9/7. Source normalization removes this obstruction and yields prediction consistency for bounded linear recursions of any finite order. We also characterize the two-lag companion systems preserving a prescribed block metric, making the corresponding self-kernel invariant along their trajectories.

1 Introduction

Causal Transformers can perform autoregressive prediction in context, and causal kernel descent (CKD) provides an explicit mechanism for this behavior [3, 4]. In the first-order setting $x_{t+1} = f(x_t)$, CKD admits prediction guarantees in several specific settings. Higher-order recurrences can be reduced to this setting by augmenting the state [3]. For example, a two-lag recurrence

$$x_{t+1} = g(x_t, x_{t-1}), \quad x_t \in \mathbb{R}^d, \quad (1)$$

becomes first order after setting

$$z_t = (x_t, x_{t-1}), \quad z_{t+1} = F(z_t) = (g(x_t, x_{t-1}), x_t). \quad (2)$$

We ask whether CKD consistency is preserved under this augmentation.

For pair states $p_t = (x_t, x_{t-1})$, direct pair-kernel CKD agrees with the corresponding block of lifted CKD. Unnormalized CKD, however, fits the scaled targets $\kappa_t x_{t+1}$, where $\kappa_t = K_2(p_t, p_t)$. If κ_t varies along the trajectory, these targets need not remain linearly representable. We give a bounded scalar linear recurrence for which the resulting fixed-point prediction error does not vanish.

Normalizing each source contribution by its self-kernel restores the original targets. The resulting fixed-point recursion is a Kaczmarz projection [1, 2]. Key-norm-normalized Kaczmarz updates also appear in linear attention [5]. A projection-energy argument gives consistency under a Hilbert–Schmidt realizability condition and uniformly bounded self-kernels. For the affine kernel, the same argument applies to bounded linear recurrences of arbitrary finite order.

For a linear recurrence $x_{t+1} = A_1 x_t + A_2 x_{t-1}$, consider the companion matrix and the family of metrics

$$C = \begin{bmatrix} A_1 & A_2 \\ I_d & 0 \end{bmatrix}, \quad G_\tau = \begin{bmatrix} I_d & \tau I_d \\ \tau I_d & I_d \end{bmatrix}, \quad |\tau| < 1. \quad (3)$$

Euclidean orthogonality of C forces $A_1 = 0$,

$$C^\top G_\tau C = G_\tau \iff A_2 \in \text{O}(d), \quad A_1 = \tau(A_2 - I_d).$$

For each fixed τ , the same metric G_τ is preserved by a nontrivial family of two-lag recurrences, so the associated quadratic self-kernel is constant along their orbits.

2 Problem and Causal Kernel Descent

The initial states x_0, x_1 are observed. At position $t \geq 1$, the input pair is $p_t = (x_t, x_{t-1})$, the lifted state is $z_t = p_t$, and the target is $y_t = x_{t+1}$. We use p_t and z_t to distinguish prediction in $\mathbb{R}^d$ from prediction in $\mathbb{R}^{2d}$ followed by projection.

Let K_2 be a real positive-semidefinite kernel on $\mathbb{R}^{2d}$, with feature map ϕ_2 in a Hilbert space $\mathcal{H}_2$. Write

$$\phi_t = \phi_2(p_t), \quad \kappa_t = K_2(p_t, p_t) > 0, \quad g(p) = W_g \phi_2(p). \quad (4)$$

The residual theorem assumes that $W_g : \mathcal{H}_2 \rightarrow \mathbb{R}^d$ is Hilbert–Schmidt. For finite-dimensional features, this is simply a matrix with finite Frobenius norm. Set

$$K_2(p, q) = p^\top q, \quad K_{2,G}(p, q) = \exp(p^\top G q), \quad G \succ 0. \quad (5)$$

For source normalization, set

$$L_{t,s} = \mathbf{1}_{s \leq t} \frac{K_2(p_t, p_s)}{\kappa_s}, \quad L_{t,t} = 1. \quad (6)$$

For a causal scalar matrix A and step size $\eta > 0$, direct CKD is

$$v_t^{\ell+1} = v_t^\ell - \eta \sum_{s \leq t} A_{t,s} (v_s^\ell - \mathbf{1}_{s < t} y_s), \quad v_t^\ell \in \mathbb{R}^d. \quad (7)$$

Lifted CKD uses the same weights and lifted targets:

$$u_t^{\ell+1} = u_t^\ell - \eta \sum_{s \leq t} A_{t,s} (u_s^\ell - \mathbf{1}_{s < t} z_{s+1}), \quad u_t^\ell \in \mathbb{R}^{2d}. \quad (8)$$

Since all target terms have $s < t$, the updates are causal. If A has positive diagonal, the finite-prefix fixed point is unique and is consistent across prefixes at common positions.

Let $\Pi_x : \mathbb{R}^{2d} \rightarrow \mathbb{R}^d$ select the first block.

Theorem 1 (Exact direct–lifted equivalence). *Run (7) and (8) with the same scalar attention matrix, step size, and depth. If $v_t^0 = \Pi_x u_t^0$ at every position, then*

$$v_t^\ell = \Pi_x u_t^\ell \quad \text{for every } t \text{ and } \ell \geq 0. \quad (9)$$

Proof. Apply Π_x to (8) and use $\Pi_x z_{s+1} = y_s$. The resulting recursion is (7); induction gives the claim. $\square$

A prepared token contains the current and previous pair, the shifted observed target, a prediction slot, boundary flags, and the required self-kernel features. For exponential kernels the latter are $\ell_s = \log \kappa_s$. These are token-wise preprocessing features.

For $K_2(p, q) = \exp(p^\top G q)$, the source-normalized score is

$$\tilde{h}_{t,s} = p_t^\top G p_s - \ell_s, \quad \tilde{h}_{t,t} = 0. \quad (10)$$

One head supplies the weighted prediction values, and a second supplies the shifted observed targets. Their signs implement the two terms in (7).

Proposition 1 (Attention realization). *On exact prepared tokens, two unnormalized exponential heads with score (10) implement one CKD update with $A = L$. With row-softmax, a zero-score, zero-value boundary source in the shifted head matches the two normalizers and implements the row-normalized matrix $\bar{L}_{t,s} = L_{t,s} / \sum_{r \leq t} L_{t,r}$. Projection of the value maps gives the direct construction from the lifted one.*

Both L and $\bar{L}$ have the same fixed point, since positive row scaling leaves a triangular fixed-point equation unchanged. Appendix A.1 gives the query, key, and value features.

3 Main Results

3.1 Source Normalization and Prediction Consistency

Lemma 1 (The fixed-point interpolation target). *Let $A_{t,s} = \mathbf{1}_{s \leq t} d_t K_2(p_t, p_s) c_s$, where $d_t, c_t > 0$. Set $r_t = y_t - v_t^*$ and $W_t = \sum_{s \leq t} c_s r_s \phi_s^*$, with $W_0 = 0$. Then*

$$\begin{aligned} v_t^* &= \frac{W_{t-1} \phi_t}{\kappa_t c_t}, & h_t &= \kappa_t c_t y_t, \\ W_t &= W_{t-1} + \frac{h_t - W_{t-1} \phi_t}{\kappa_t} \phi_t^*. \end{aligned} \tag{11}$$

Proof. The t th fixed-point equation, divided by d_t , reads $\kappa_t c_t v_t^* = \sum_{s < t} K_2(p_t, p_s) c_s r_s = W_{t-1} \phi_t$. Subtract from $\kappa_t c_t y_t$ and substitute in $W_t - W_{t-1}$. $\square$

Unnormalized CKD has $c_t = 1$ and fits $h_t = \kappa_t y_t$. Source normalization has $c_t = 1/\kappa_t$ and fits $h_t = y_t$. When the self-kernel varies, these targets can have different realizability properties even on a bounded orbit.

Proposition 2 (A bounded AR(2) counterexample). *Let $x_0 = 1$, $x_1 = 2$, and $x_{t+1} = -x_t - x_{t-1}$. For the unnormalized linear kernel $A_{t,s} = \mathbf{1}_{s \leq t} p_t^\top p_s$, the depth iteration converges at every fixed prefix with $\eta = 1/13$. Its fixed-point residual satisfies*

$$4r_{3m+2} + 10r_{3m+3} = 18 \quad (m \geq 0), \quad \lim_{t \rightarrow \infty} r_t = \frac{9}{7}. \tag{12}$$

The three inputs are $(2, 1)$, $(-3, 2)$, and $(1, -3)$, with self-kernels 5, 13, 10. Their sum is zero, whereas the scaled targets $-15, 13, 20$ sum to 18. A linear function cannot interpolate those scaled targets. The first identity in (12) already gives a residual of magnitude at least $9/7$ in every three-position block. Appendix A.2 proves the identity and the exact limit.

Theorem 2 (Direct pair-kernel residual budget). *Assume (4) with W_g Hilbert–Schmidt. The fixed point of (7) with $A = L$ satisfies, for every N ,*

$$\sum_{t=1}^N \frac{\|y_t - v_t^*\|^2}{\kappa_t} = \|W_g\|_{\text{HS}}^2 - \|W_g - W_N\|_{\text{HS}}^2 \leq \|W_g\|_{\text{HS}}^2, \tag{13}$$

where $W_N = \sum_{s \leq N} (r_s / \kappa_s) \phi_s^*$. If $\kappa_t \leq \kappa_{\max} < \infty$, then

$$\sum_{t \geq 1} \|y_t - v_t^*\|^2 \leq \kappa_{\max} \|W_g\|_{\text{HS}}^2, \quad \|y_t - v_t^*\| \rightarrow 0. \tag{14}$$

Proof. Lemma 1 gives $v_t^* = W_{t-1}\phi_t$. Set $E_t = W_g - W_t$, $\nu_t = \phi_t/\sqrt{\kappa_t}$, and $P_t = I - \nu_t\nu_t^*$. Then $E_t = E_{t-1}P_t$ and

$$\|E_{t-1}\|_{\text{HS}}^2 - \|E_t\|_{\text{HS}}^2 = \|E_{t-1}\nu_t\|^2 = \frac{\|r_t\|^2}{\kappa_t}.$$

Summation proves (13). The self-kernel bound gives (14). In finite dimension these are ordinary Frobenius norm identities. $\square$

Corollary 1 (Bounded linear recursions of finite order). *Let $x_{t+1} = B\zeta_t = \sum_{j=1}^p A_j x_{t-j+1}$, where $\zeta_t = (x_t, \dots, x_{t-p+1})$ and $\sup_t \|\zeta_t\| \leq R$. For source-normalized CKD, the kernel $K(\zeta, \xi) = 1 + \zeta^\top \xi$ gives*

$$\sum_{t \geq 1} \|y_t - v_t^*\|^2 \leq (1 + R^2) \sum_{j=1}^p \|A_j\|_F^2. \quad (15)$$

The affine feature $(1, \zeta)$ has representing matrix $(0, B)$ and self-kernel at most $1 + R^2$. Theorem 2 gives the bound.

Proposition 3 (Fixed-context depth convergence). *At prefix length T , let A be lower triangular with positive diagonal. If $\max_{t \leq T} |1 - \eta A_{t,t}| < 1$, then the depth iteration converges to its unique fixed point. For $A = L$, every $0 < \eta < 2$ suffices, and at $\eta = 1$ the iterate equals the fixed point after $n \geq T$ layers.*

Indeed, the error is $(I - \eta A)^n$ times the initial error. For $A = L$ and $\eta = 1$, the iteration matrix is strictly lower triangular and its T th power is zero. Combining this observation with Corollary 1 gives $\|v_T^{(T)} - x_{T+1}\| \rightarrow 0$. The row-normalized matrix has the same fixed point and positive diagonal at most one; its depth convergence also holds for $0 < \eta < 2$.

3.2 Companion Geometry and Shared Construction

Theorem 3 (Companion isometries of G_τ). *Fix $\tau \in (-1, 1)$ and the matrices in (3). Then*

$$C^\top G_\tau C = G_\tau \iff A_2 \in \text{O}(d), \quad A_1 = \tau(A_2 - I_d). \quad (16)$$

The family has nonzero coefficients on both lags when $\tau \neq 0$ and $A_2 \neq I_d$. For fixed τ , the metric G_τ is independent of A_2 .

For the counterexample, $A_1 = A_2 = -1$ and $\tau = 1/2$, so $p_t^\top G_{1/2} p_t = 7$. The Euclidean self-kernel varies and produces the incompatible scaled targets in Proposition 2; the invariant metric makes the scale constant.

Corollary 2 (Fixed- τ causal realization). *Fix $d \geq 1$ and $0 < |\tau| < 1$. For each $Q \in \text{O}(d)$ let $A_2 = Q$ and $A_1 = \tau(Q - I_d)$. For a nonzero initial pair, normalize the prompt by $s_\tau = (z_1^\top G_\tau z_1)^{1/2}$ and write $\bar{x}_t = x_t/s_\tau$. There is a prepared-token attention construction whose parameters depend on (d, τ) , but not on Q , such that its exact-token, source-normalized iterate with step size $\eta = 1$ satisfies*

$$\lim_{T \rightarrow \infty} \|v_T^{(T)} - \bar{x}_{T+1}\| = 0. \quad (17)$$

The normalized trajectory is G_τ -unit, so the metric-exponential self-kernel is the constant e . Its linear target has a finite-norm representation (Appendix A.3). Theorem 2 gives residual convergence, Proposition 1 realizes the update, and Proposition 3 reaches the fixed point in T layers.

4 Numerical Illustration

On the AR(2) orbit of Proposition 2, Figure 1 compares unnormalized Euclidean linear CKD, its source-normalized version, and unnormalized linear CKD with $G_{1/2}$. The unnormalized Euclidean predictor retains the 9/7 residual, while both corrections reach numerical precision. The identity (12) is satisfied to 7.2×10^{-15} ; see Appendix A.4 for numerical details.

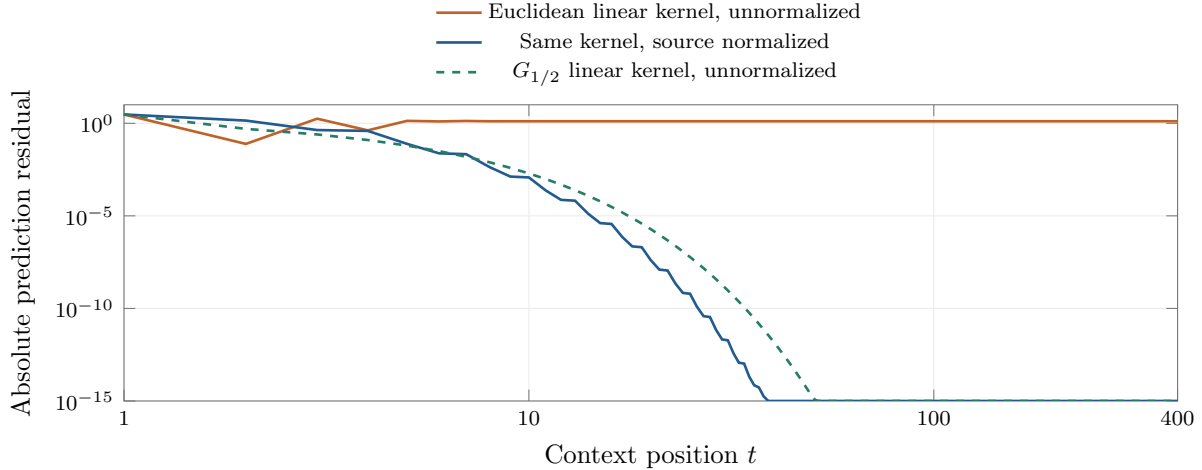


Figure 1: Fixed-point prediction residuals on the AR(2) orbit of Proposition 2, with $T = 400$. Values below 10^{-15} are displayed at the plotting floor.

References

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A Proofs and Experimental Details

A.1 Query, key, and value maps

Write $G = R^\top R$. The residual head has

$$q_t^- = (Rp_t, 1), \quad k_s^- = (Rp_s, -\ell_s), \quad a_s^- = -\eta v_s^\ell.$$

For the shifted-target head, use the same query, and for $s > 1$ set

$$k_s^+ = (Rp_{s-1}, -\ell_{s-1}), \quad a_s^+ = \eta x_s.$$

At $s = 1$, the shifted value is zero and the key has score zero. With unnormalized exponentials the two sums are

$$-\eta \sum_{s \leq t} L_{t,s} v_s^\ell + \eta \sum_{s < t} L_{t,s} y_s,$$

which gives (7) after adding the residual connection. For softmax the denominators are respectively $\sum_{s \leq t} e^{\tilde{h}_{t,s}}$ and $1 + \sum_{s < t} e^{\tilde{h}_{t,s}}$. They agree because $\tilde{h}_{t,t} = 0$. Using u_s^ℓ and z_s as values gives the lifted construction.

A.2 The AR(2) counterexample

One period has inputs $a = (2, 1)$, $b = (-3, 2)$, $c = (1, -3)$ and targets $-3, 1, 2$. The inner products are $\|a\|^2 = 5$, $\|b\|^2 = 13$, $\|c\|^2 = 10$, $a^\top b = -4$, $a^\top c = -1$, $b^\top c = -9$. At fixed prefix length, the diagonal entries of the depth error matrix $I - A/13$ are $8/13, 0, 3/13$, so its spectral radius is less than one.

Let w be the online parameter before a period, with residuals $\rho_i = r_{3m+i}$. Lemma 1 gives

$$\begin{aligned} w^\top a &= -15 - 5\rho_1, \\ w^\top b &= 13 + 4\rho_1 - 13\rho_2, \\ w^\top c &= 20 + \rho_1 + 9\rho_2 - 10\rho_3. \end{aligned}$$

Since $a + b + c = 0$, summation gives $4\rho_2 + 10\rho_3 = 18$.

For the exact limit, write each parameter step as $w_i = P_i w_{i-1} + y_i p_i$, where $P_i = I - p_i p_i^\top / \kappa_i$. The linear part of a full period is $P_3 P_2 P_1$, and $\|P_3 P_2 P_1\| \leq \|P_2 P_1\| = 4/\sqrt{65} < 1$. The period map is therefore an affine contraction. At its unique periodic steady state, the net update is $\rho_1 a + \rho_2 b + \rho_3 c = 0$. As the only relation among a, b, c has equal coefficients, $\rho_1 = \rho_2 = \rho_3 = \rho$. Hence $14\rho = 18$, proving the limit.

A.3 Companion geometry

The eigenvalues of G_τ are $1 + \tau$ and $1 - \tau$, each repeated d times. Direct multiplication gives

$$C^\top G_\tau C = \begin{bmatrix} A_1^\top A_1 + \tau(A_1^\top + A_1) + I_d & (A_1^\top + \tau I_d) A_2 \\ A_2^\top (A_1 + \tau I_d) & A_2^\top A_2 \end{bmatrix}. \quad (18)$$

Equality with G_τ first gives $A_2^\top A_2 = I_d$. The off-diagonal equation then gives $A_1 = \tau(A_2 - I_d)$. Conversely, substituting $A_2 = Q$ and $A_1 = \tau(Q - I_d)$ in (18) returns G_τ .

Along the normalized orbit, $p_t^\top G_\tau p_t = 1$. With $R = G_\tau^{1/2}$, the metric-exponential kernel has feature map

$$\phi_G(p) = \bigoplus_{m \geq 0} \frac{(Rp)^{\otimes m}}{\sqrt{m!}}.$$

Its degree-one block is Rp . The target matrix $B_Q = [\tau(Q - I_d) \ Q]$ is represented on that block by $B_Q R^{-1}$, with finite Frobenius norm. Thus Theorem 2 applies with $\kappa_t = e$.

A.4 Experimental protocol

Core comparison. The scalar trajectory in Proposition 2 is evaluated for $T = 400$ prediction positions. All online parameters start at zero. For source-normalized linear CKD the update is $v_t^* = W_{t-1}p_t$, $W_t = W_{t-1} + r_t p_t^\top / \|p_t\|^2$. For unnormalized linear CKD it is $v_t^* = W_{t-1}p_t / \|p_t\|^2$, $W_t = W_{t-1} + r_t p_t^\top$. The metric control applies the latter rule to Rp_t with $R^\top R = G_{1/2}$. The target row is $[-1, -1]$. The residual-energy identity is checked at each source-normalized update. The fixed-point equations are independently checked by a triangular linear solve on the first 24 positions. The 9/7 steady cycle is also checked with rational arithmetic.

Implementation checks. Table 1 reports the largest numerical discrepancies for the shared-metric construction. The first two columns verify the relative isometry defect $\|C^\top G_\tau C - G_\tau\|_F / \|G_\tau\|_F$ and the orbit energy error $\max_{t \leq 200} |p_t^\top G_\tau p_t - 1|$. The remaining columns check direct-lifted outputs, the two explicit attention heads, and the source-normalized depth- T iterate against its triangular fixed point. These checks use both unnormalized exponential and matched row-softmax heads where applicable.

Table 1: Maximum discrepancies over 16 prompts per τ . Construction checks use prefixes of length 12. The depth column uses source-normalized exponential CKD at $\eta = 1$.

τ	Isometry	Orbit energy	Direct/lifted	Two heads	Depth T
0.2	4.0×10^{-16}	5.2×10^{-14}	0	1.1×10^{-16}	2.0×10^{-15}
0.4	5.1×10^{-16}	5.3×10^{-14}	0	1.1×10^{-16}	9.4×10^{-16}
0.7	5.5×10^{-16}	5.1×10^{-14}	0	1.1×10^{-16}	9.4×10^{-16}

We use $d = 4$ and NumPy’s default random generator with seed 20260919. For each $\tau \in \{0.2, 0.4, 0.7\}$ in increasing order, 16 Gaussian 4×4 matrices undergo QR factorization with positive diagonal correction, producing orthogonal Q . Each Gaussian initial pair is scaled to unit G_τ norm. The orbit energy is checked for 200 positions. Construction checks use the first 12 positions, zero predictions, eight layers, and $\eta = 0.25$. Both exponential and row-softmax heads use the source-corrected scores, with a matched boundary score of zero. Direct and lifted updates use the same scalar matrix. The depth-termination check instead uses unnormalized exponential heads, $\eta = 1$, and 12 layers, compared with a direct triangular solve. All errors in the last three columns of Table 1 are maximum absolute coordinate discrepancies.

The source package includes the verification script and all plotted data.